

Chapter 3: Antenna Fundamentals

Outline



Introduction



Applications of
Antennas



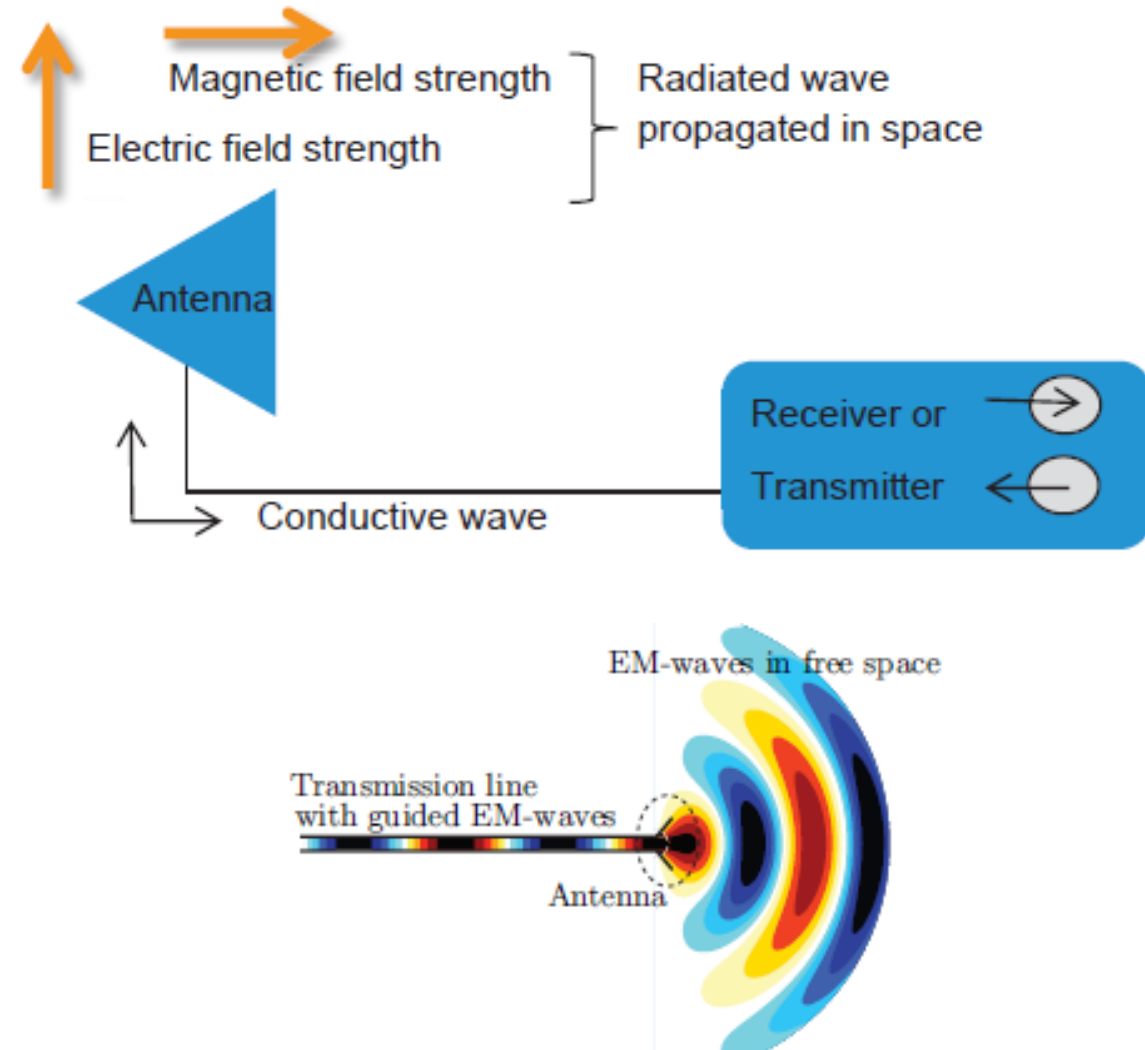
Properties of
Antennas



Types of
Antennas

Introduction

- An **antenna** (or **aerial**) is an electrical device which converts **electric power** into **radio waves**, and **vice versa**.
 - ✓ **Transmission** - radiates electromagnetic energy into space
 - ✓ **Reception** - collects electromagnetic energy from space
- In **two-way communication**, the same antenna can be used for **transmission** and **reception**
- Antennas act as **transformers** between conducted waves and electromagnetic waves propagating freely in space. So, it is basically a **transducer**.



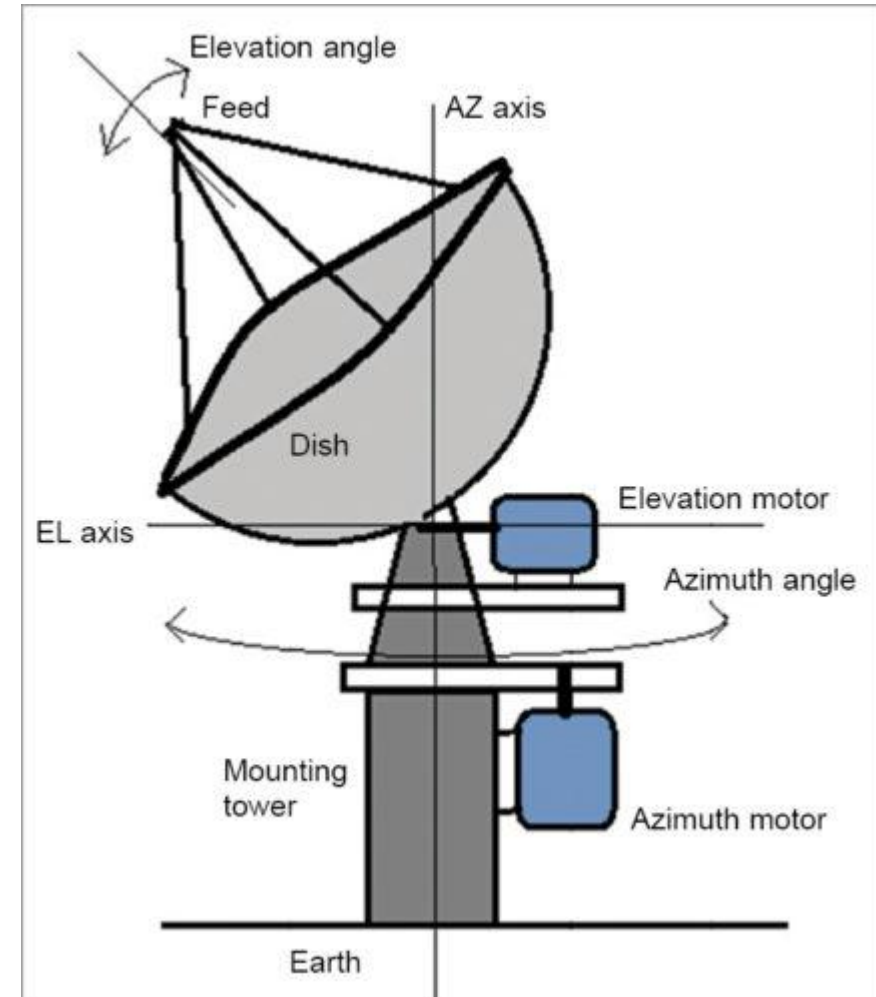
Basic Structure

Conductive Element: The core component (wire, rod, or patch) that carries current to emit or receive radiation.

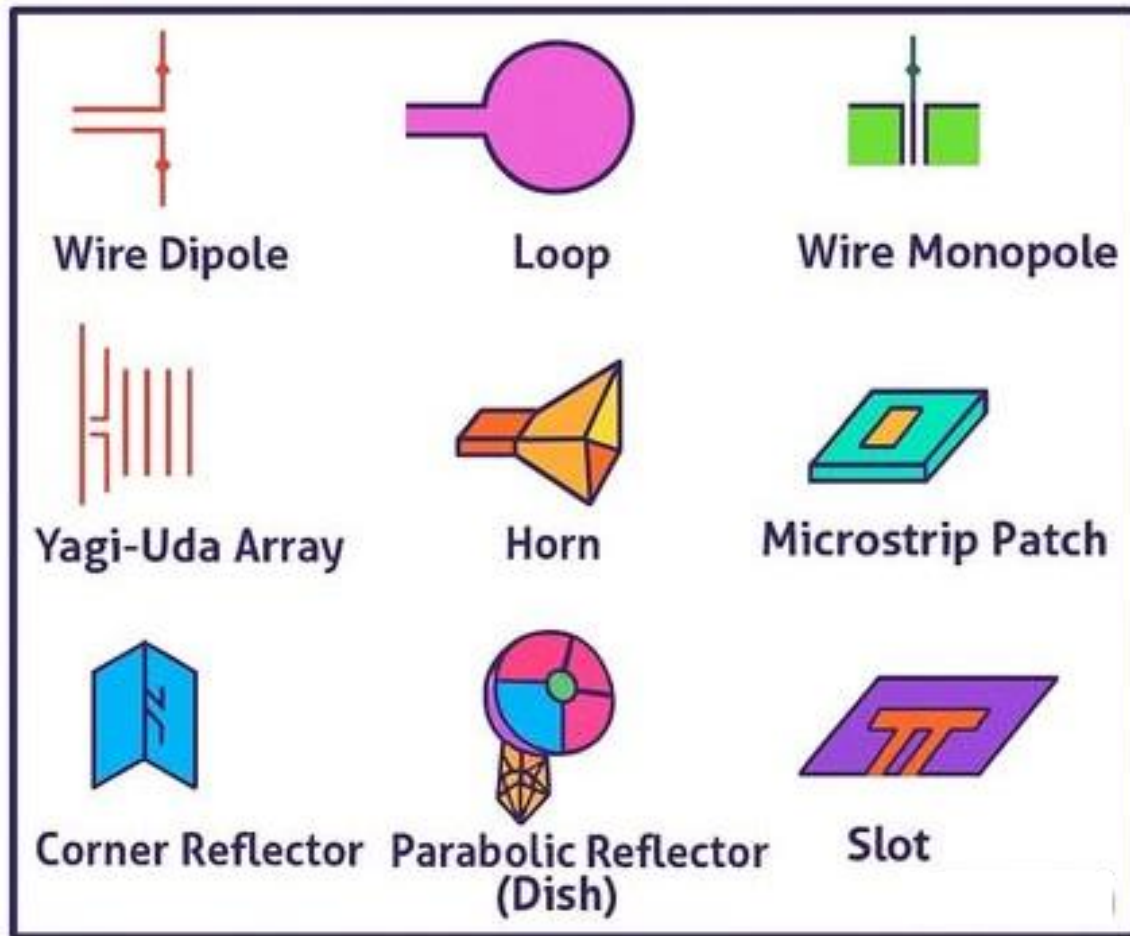
Feeding Point (Feedline): The point where a transmission line (like a coaxial cable) connects to the antenna, providing the signal.

Ground Plane / Reflector: A conductive surface that acts as a mirror for electromagnetic waves, helping to shape the antenna's beam (e.g., in monopole or parabolic antennas).

Dielectric Substrate: In patch antennas, this is a non-conductive layer between the radiator and the ground plane.



Applications



Dipole Antennas: Radio broadcasting, walkie-talkies, base stations

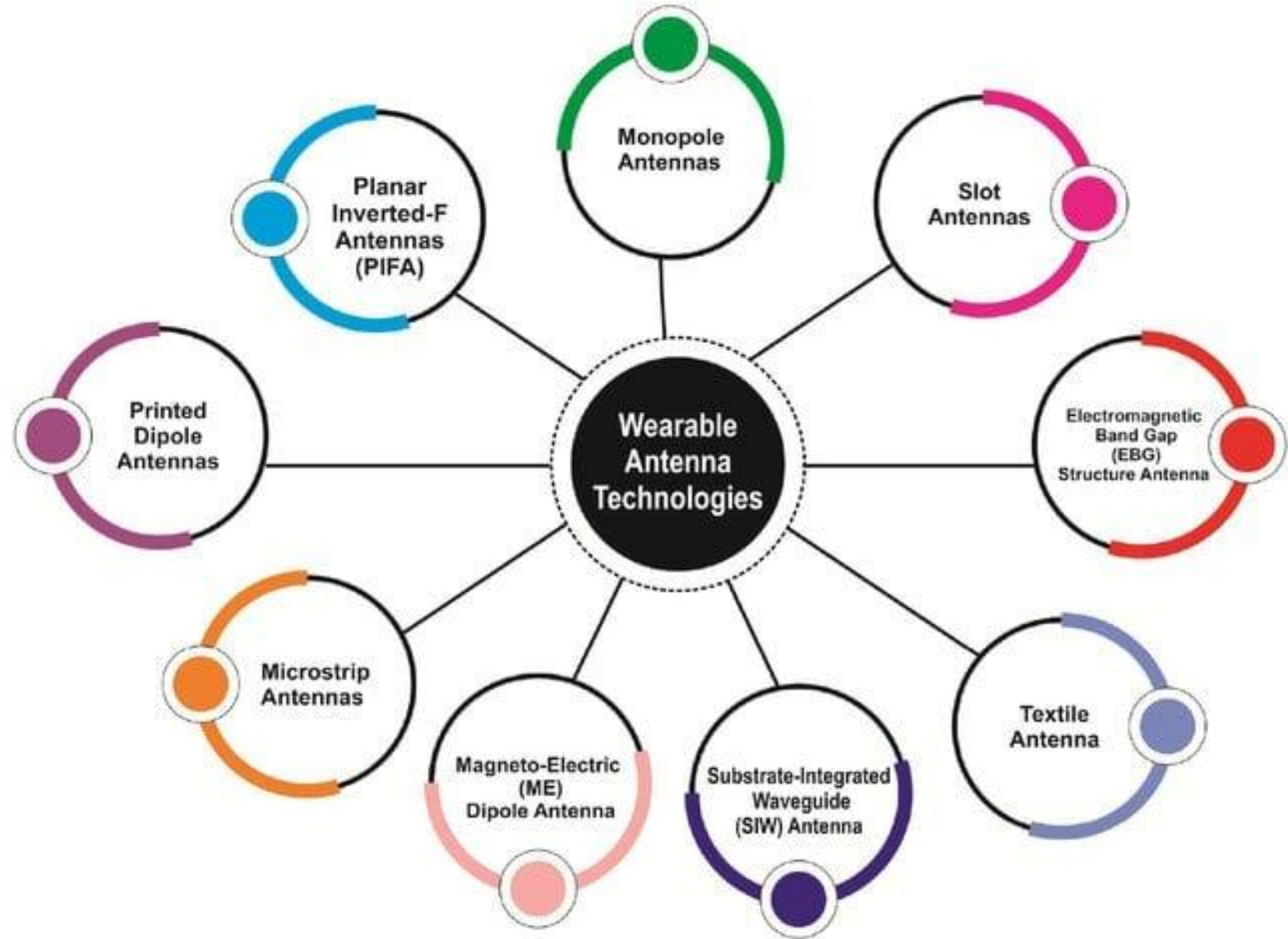
Monopole Antennas: Vehicles, mobile phones

Parabolic (Dish): Satellite TV, radar, long-distance communication

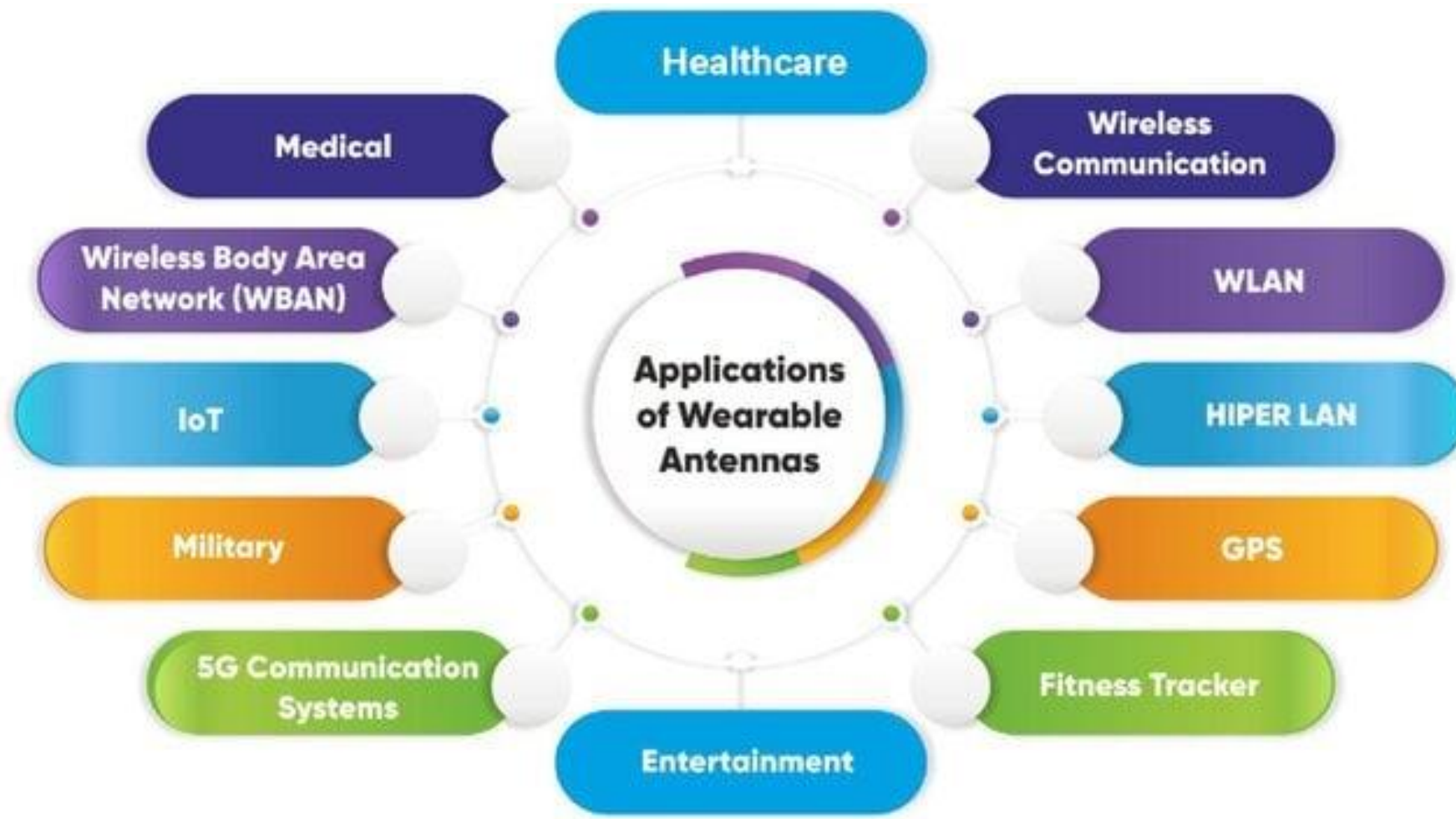
Patch (Microstrip): Mobile devices, laptops, IoT sensors

Horn Antennas: High-frequency systems, satellite communication, radar

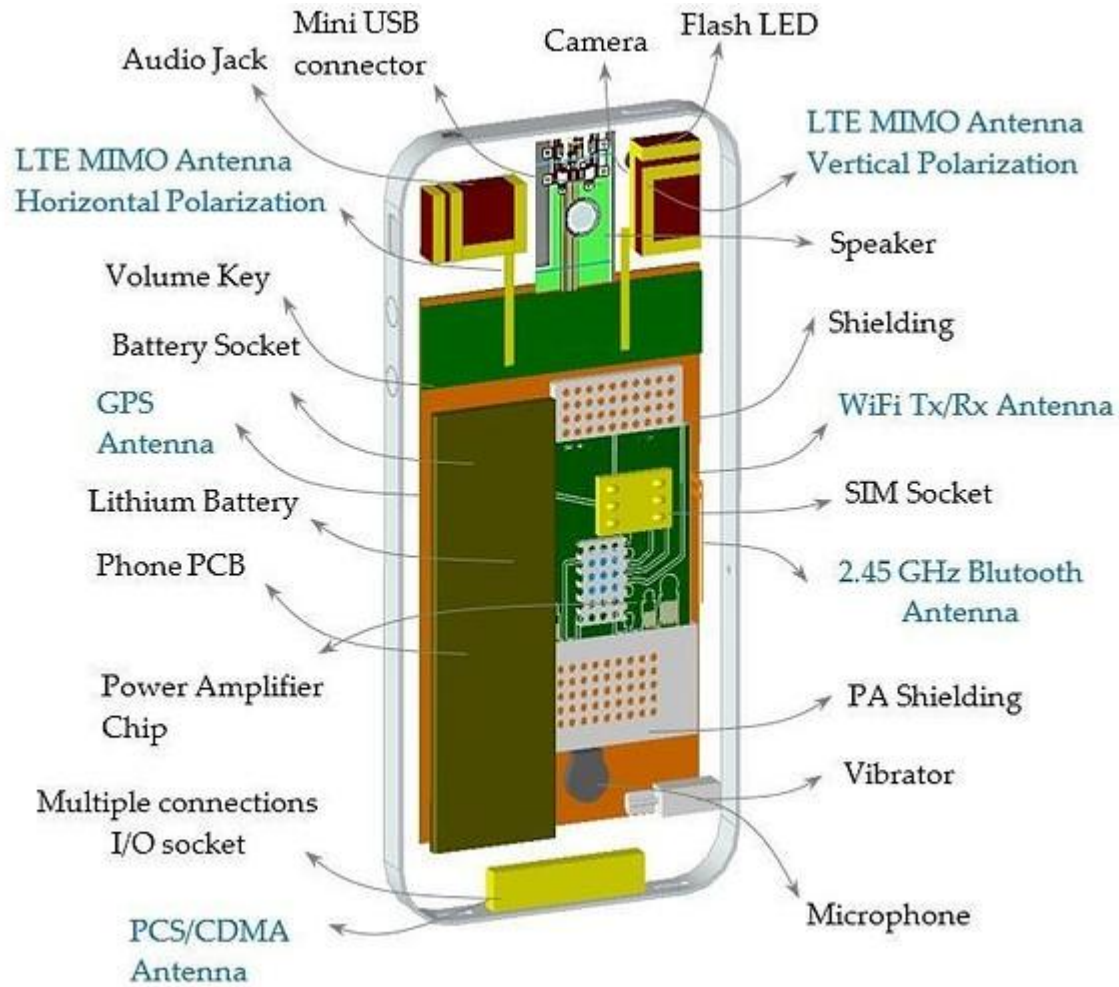
Examples



Examples



Examples



Wireless Communications Considerations



Base stations:

- High efficiency
- Directivity

Terminal antennas:

- Many small antennas in modern mobile phones.
- MIMO (many antennas), RFID, TV...
- Co-exist with camera, battery, ..., user
- Must be cost efficient.

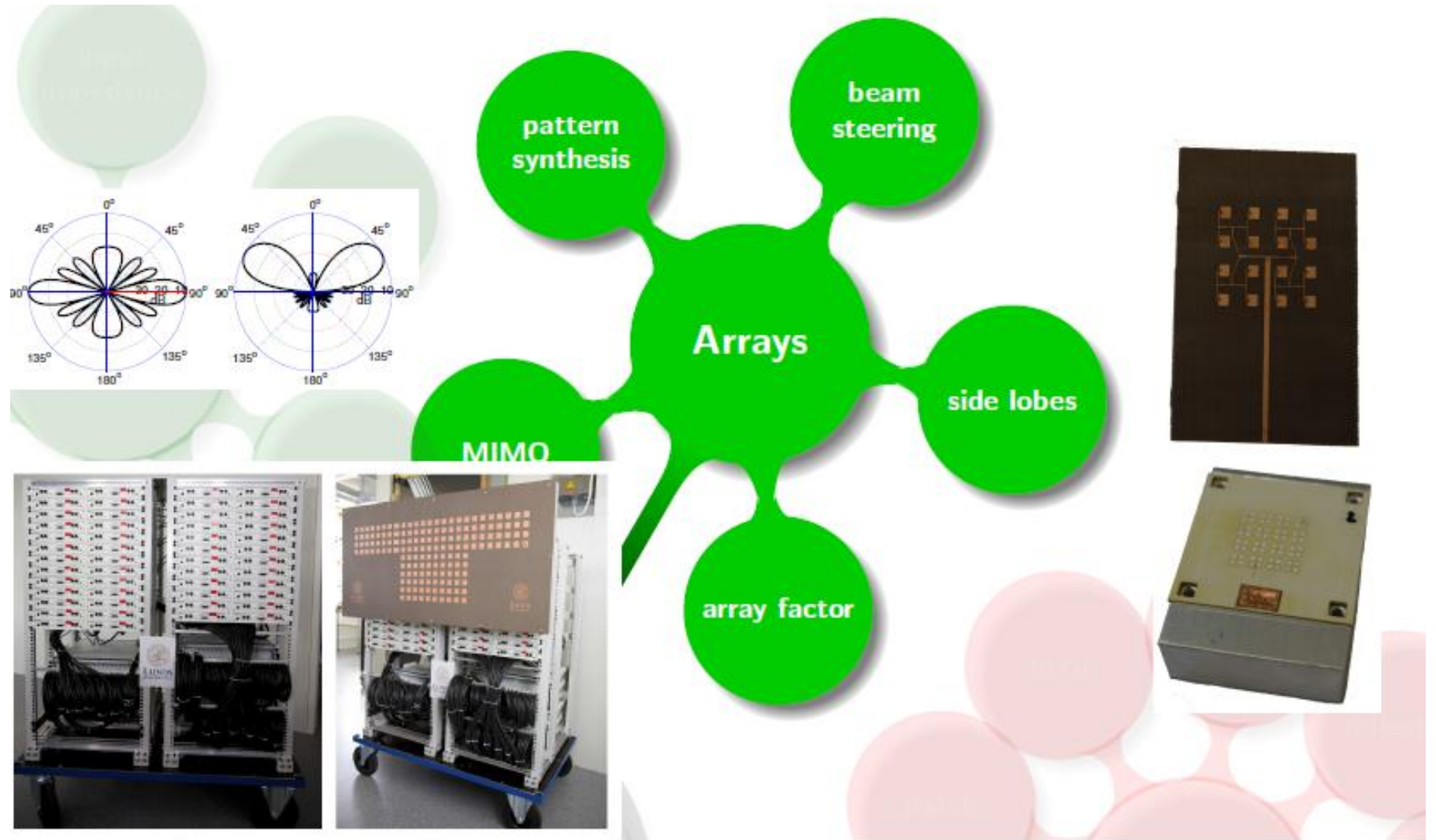
Antenna Technology



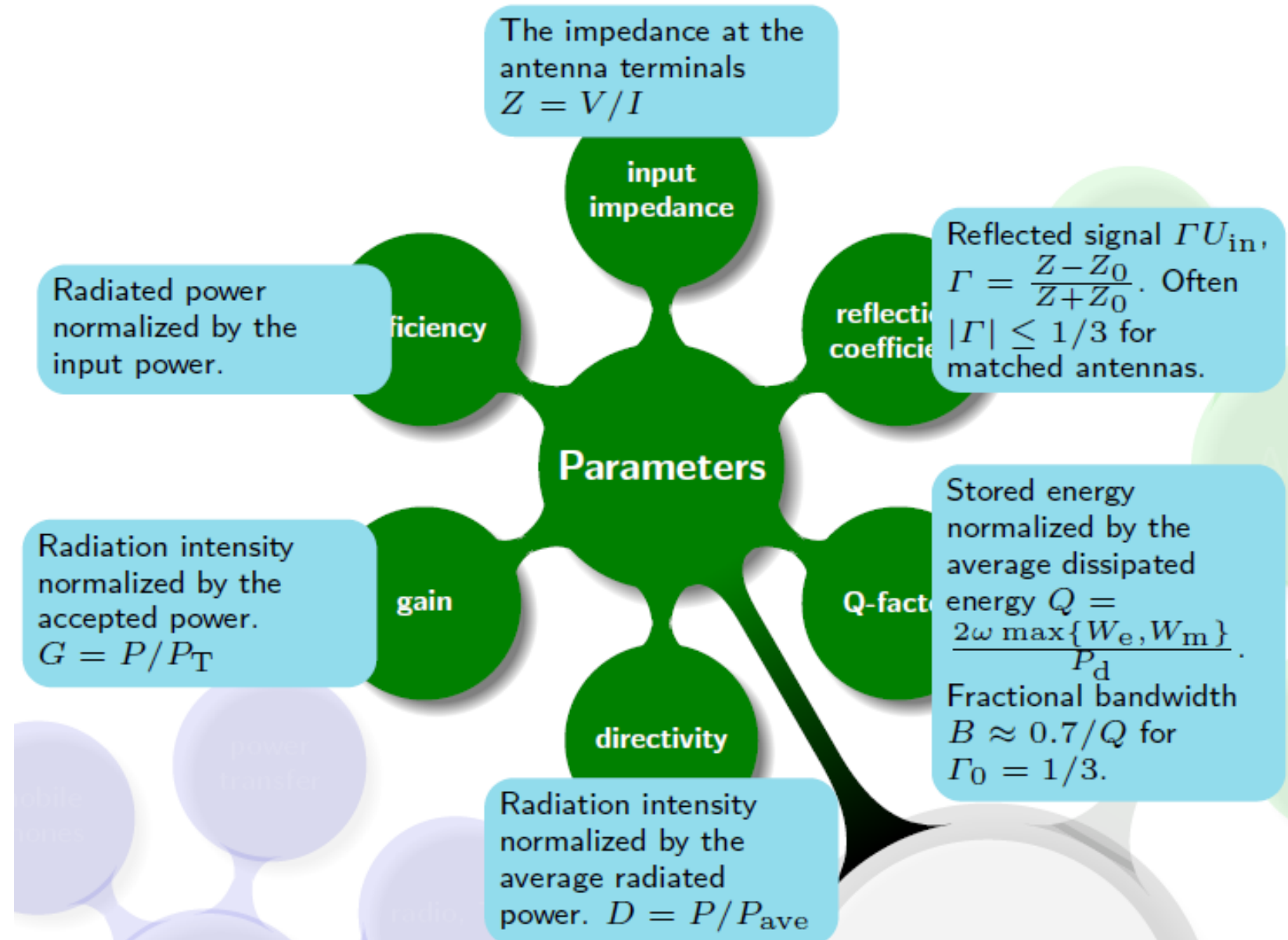
Antenna Elements



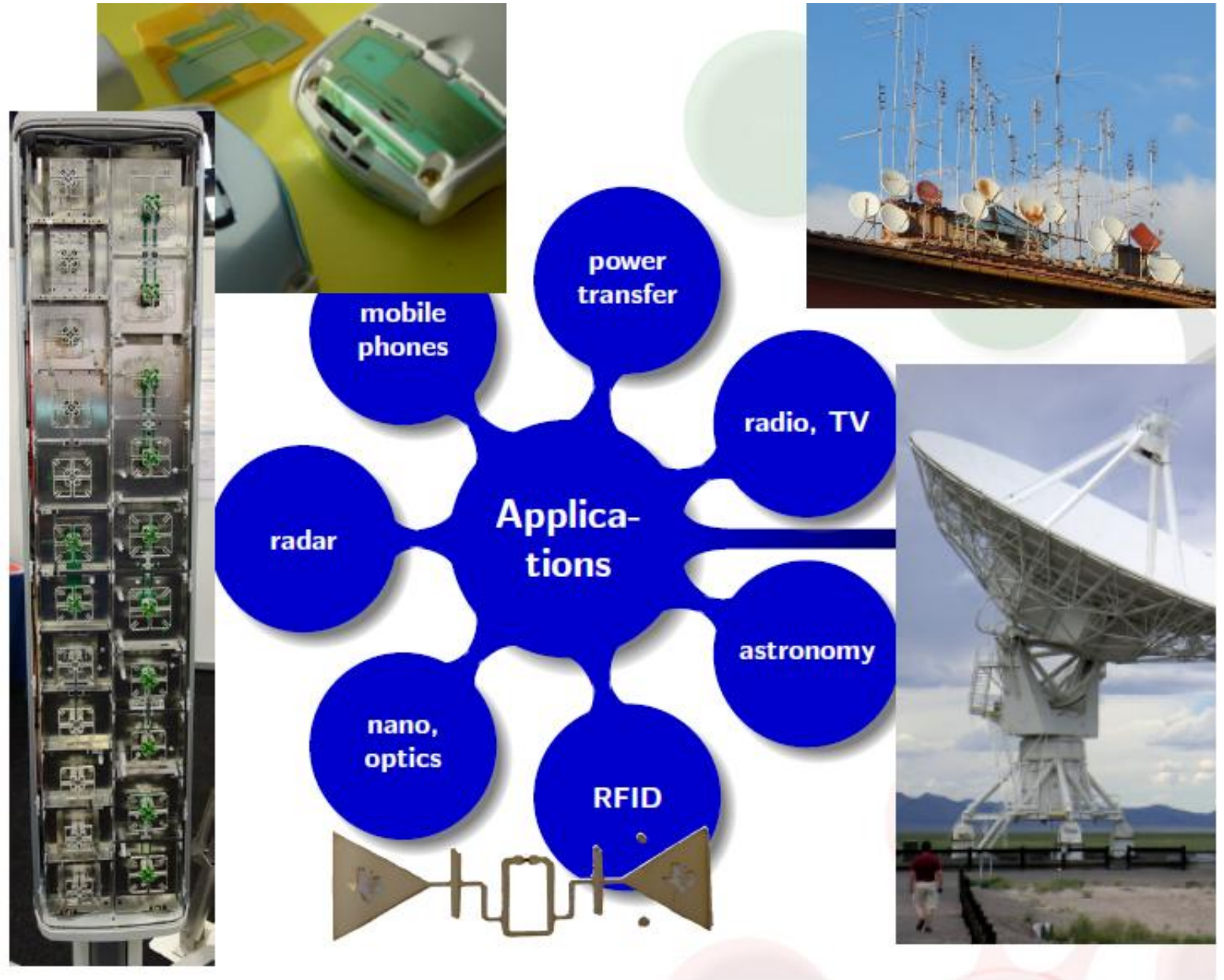
Antenna Arrays



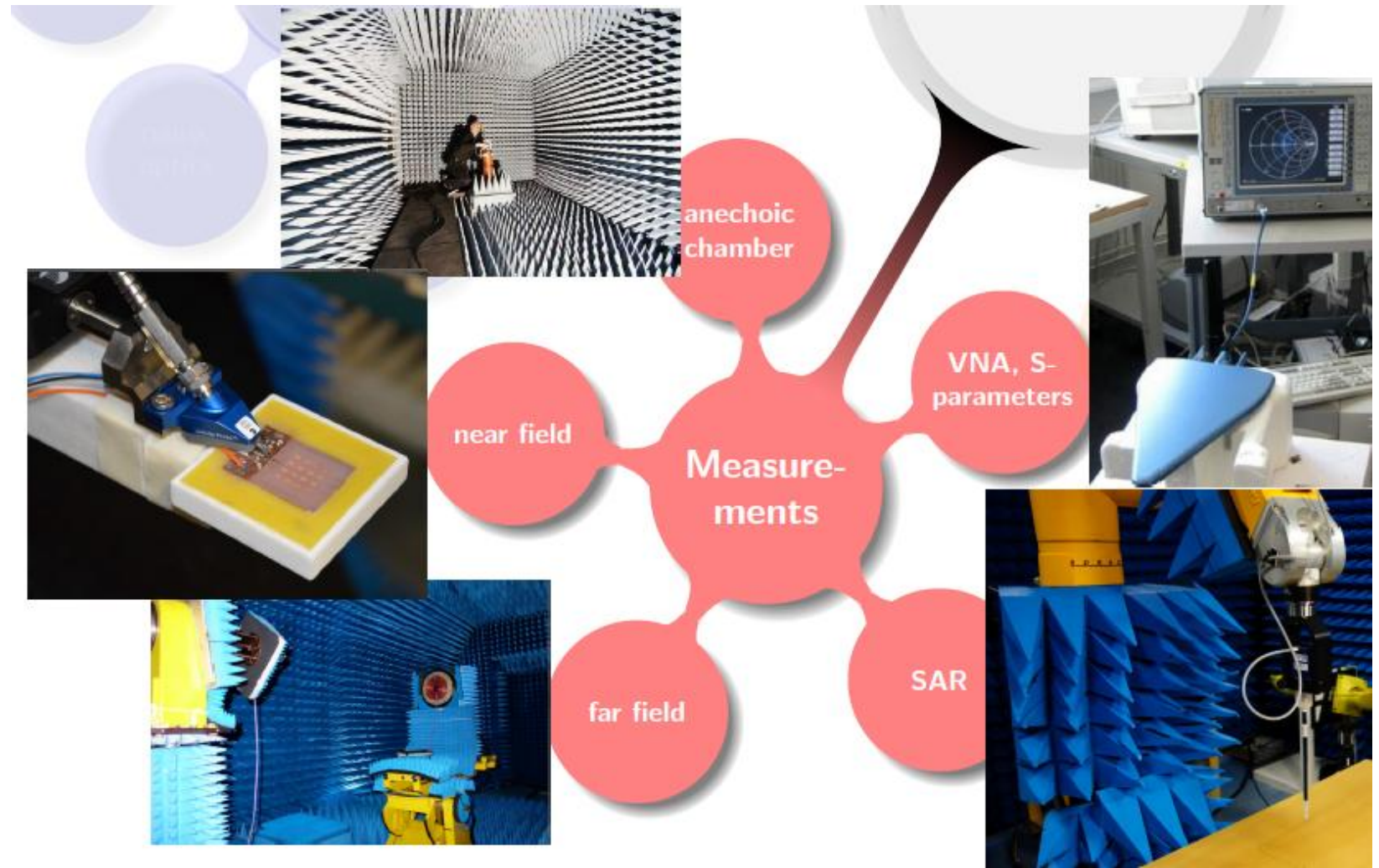
Antenna Parameters



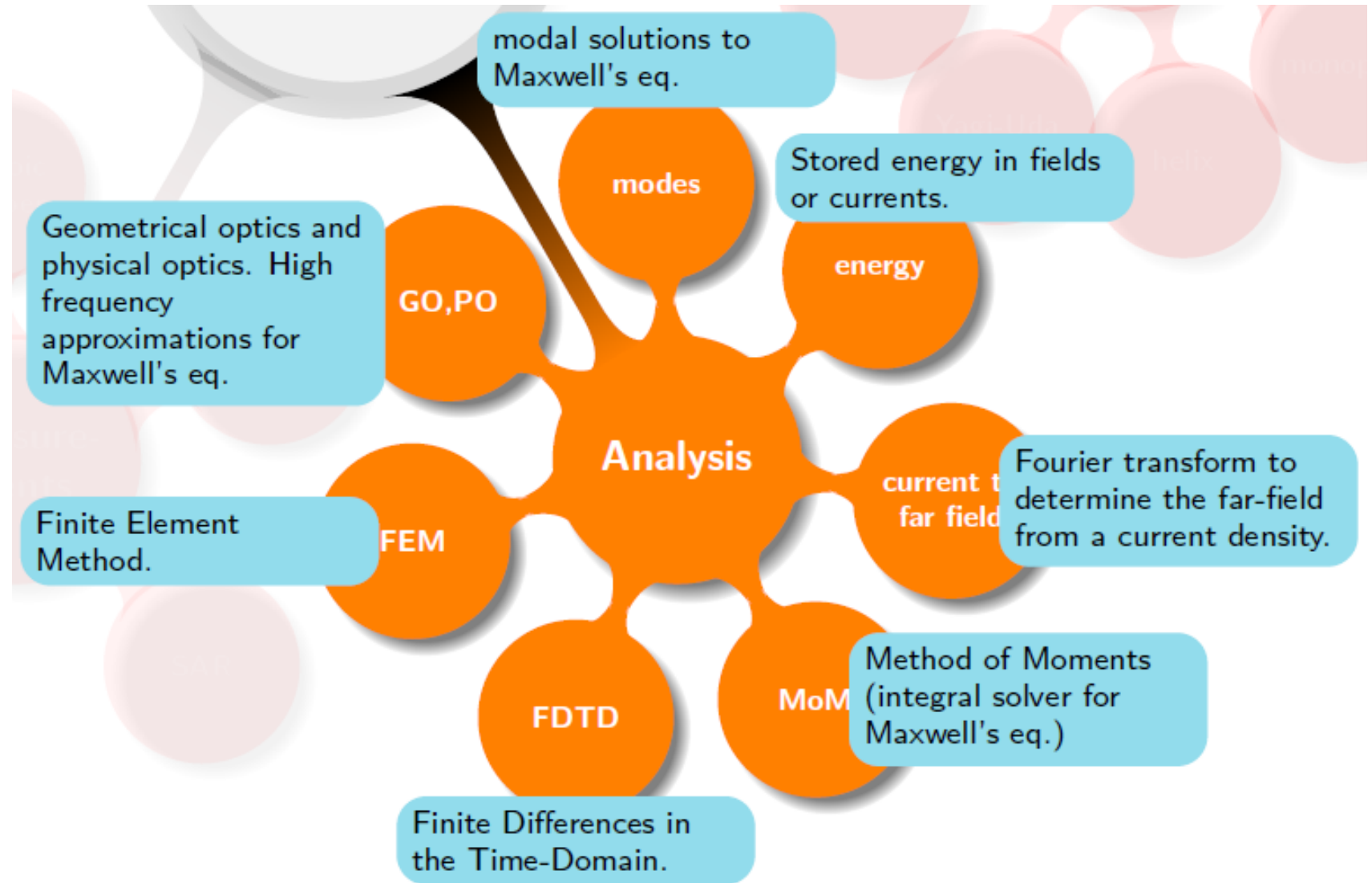
Antenna Applications



Antenna Measurements

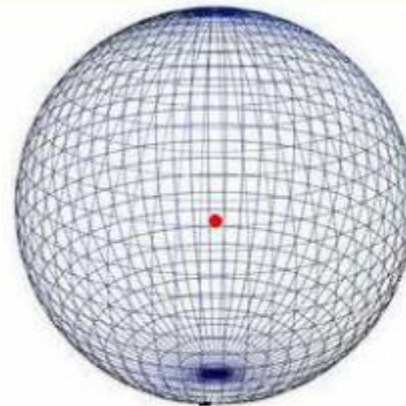


Methods of Analysis

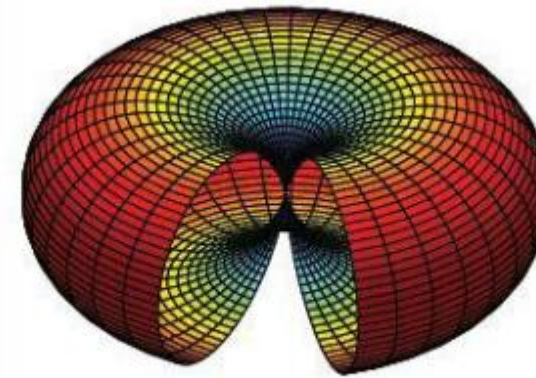


Antenna Categories

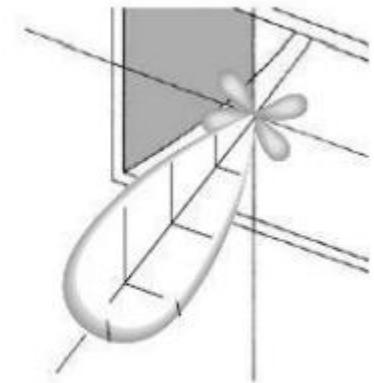
1. **Isotropic:** A hypothetical, lossless antenna has equal radiation in all direction. Not realized, but used as a reference.
2. **Omnidirectional** - Receive or radiate more or less in **all directions**.
 - These are employed when the relative position of the other station is unknown or arbitrary.
 - They are also used at lower frequencies where a directional antenna would be too large, or simply to cut costs in applications where a directional antenna isn't required.
3. **Directional** or *beam* antennas which are intended to preferentially **radiate** or **receive** in a particular **direction** or directional **pattern**.



Isotropic



Omni-directional



directional

Antenna Properties/Characteristics

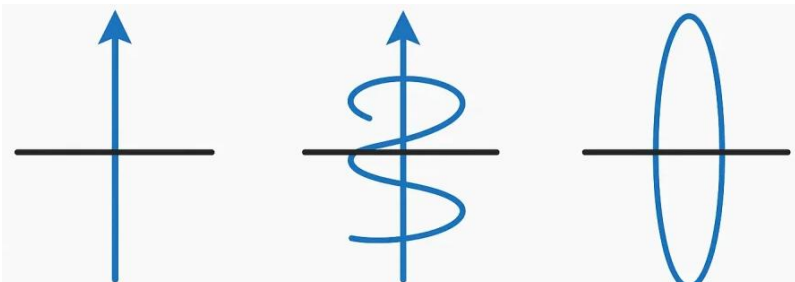
1. Reciprocity

- In general, the various properties of an antenna apply equally regardless of whether it is used for transmitting or receiving
 - ✓ If an antenna can efficiently transmit a signal in a given direction, it will also receive signals from that same direction with equal effectiveness.

- Here, the antenna exhibits identical impedance during Transmission or Reception, same directional patterns during Transmission or Reception, same effective height while transmitting or receiving.
 - ✓ In this case, transmission and reception antennas can be used interchangeably.
 - Transmission/reception efficiency
 - Gain
 - Current and voltage distribution
 - Impedance

2. Polarization

- The polarization of an antenna is determined by the direction of the electric field $\vec{E}$.
- **Linear polarization:** The $\vec{E}$ field vector changes in magnitude only. Electric field is perpendicular to the Earth's surface
 - ✓ e.g., Broadcast tower for AM radio, “whip” antenna on an automobile
 - ✓ **Horizontally Polarized Antenna** - Electric field is parallel to the Earth's surface. e.g., Television transmission (U.S.)
 - ✓ **Vertically Polarized Antenna** - Electric field is perpendicular to the Earth's surface.
- **Circular polarization:** The magnitude of the $\vec{E}$ field vector is constant, but the direction changes and rotates around the direction of propagation.
- **Elliptical polarization:** The magnitude and the direction of the $\vec{E}$ field vector changes and its peak position can be described by an elliptical equation.



Antenna polarization

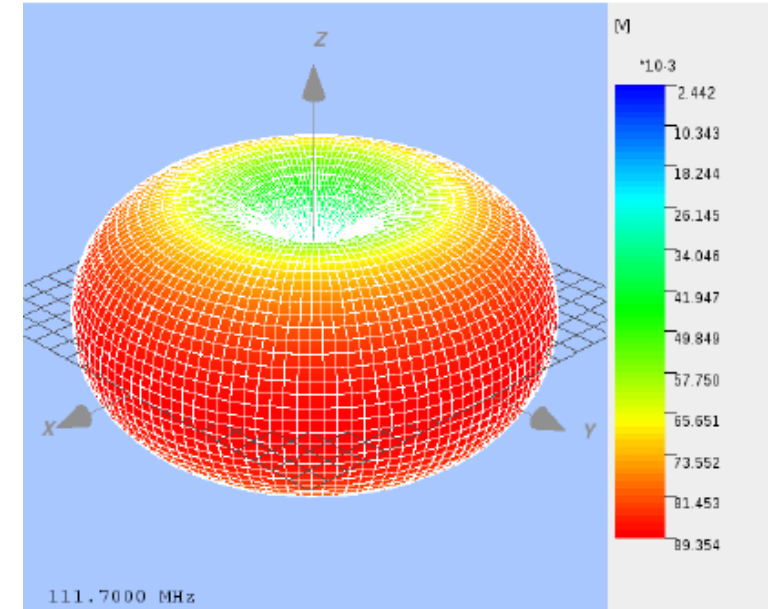
	Linear	Circular	Elliptical
Linear	0 dB	∞	3 dB
Circular	3 dB	0 dB	3 dB
Elliptical	3 dB	3 dB	0 dB

E vector of incoming signal

	Linear	Circular	Elliptical
Linear	0 dB	∞	3 dB
Circular	3 dB	0 dB	3 dB
Elliptical	3 dB	3 dB	0 dB

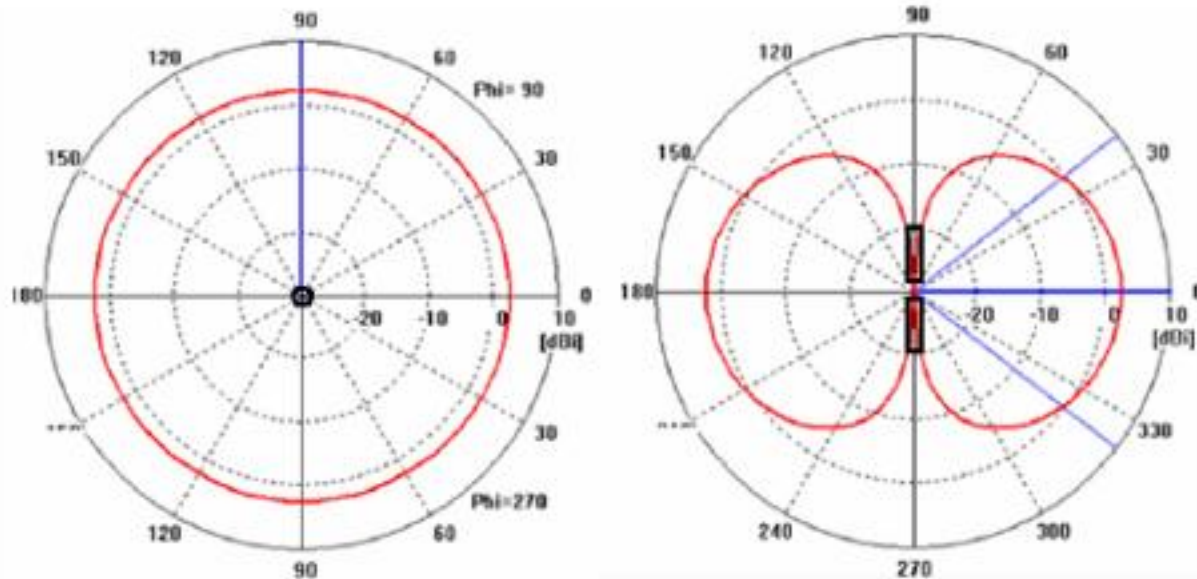
3. Radiation Pattern

- The **three-dimensional** radiation behavior of antennas is described by their **radiation pattern** (normally in the far field).
- ✓ It describes how an antenna distributes energy in space—showing the strength of the signal in different directions.



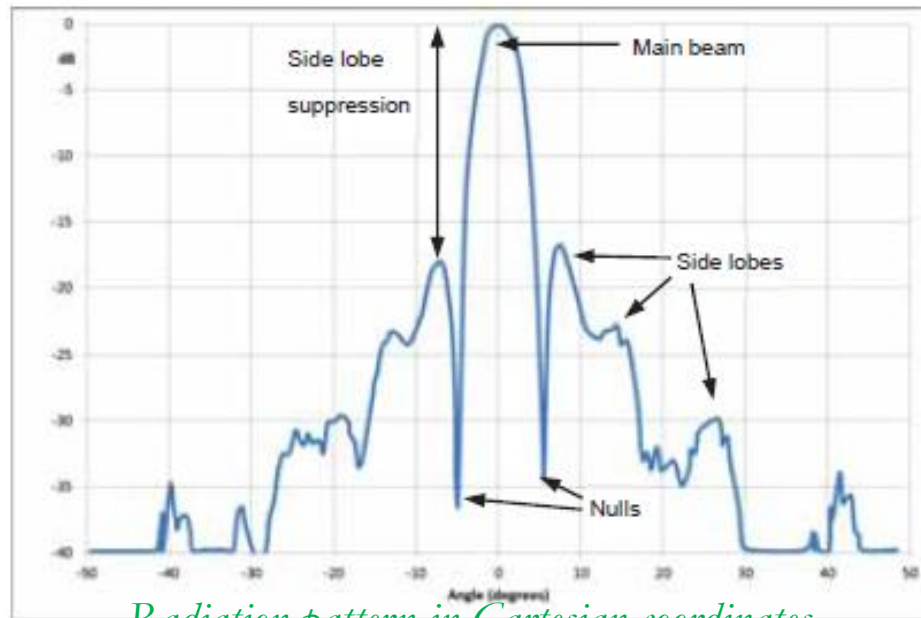
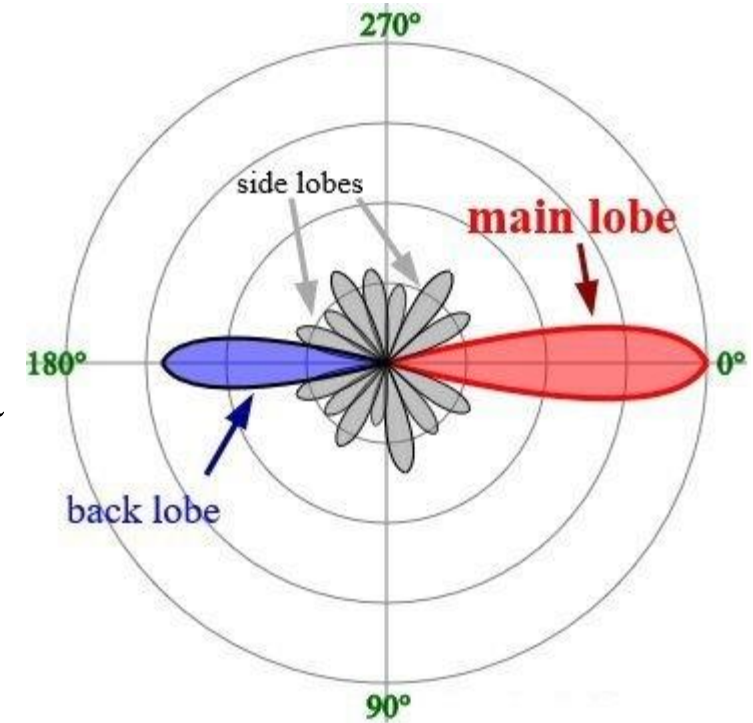
- ✓ It's essentially a directional map of an antenna's performance:
 - Where the signal is strongest
 - Where it is weak or zero

- **Horizontal pattern** or **vertical pattern** to shows the field strength.

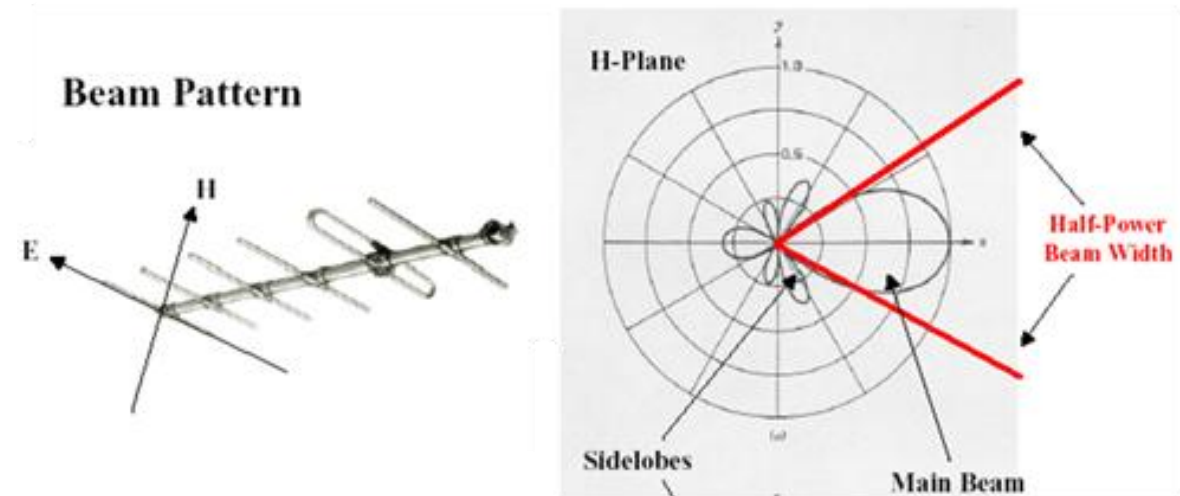


Beam Width

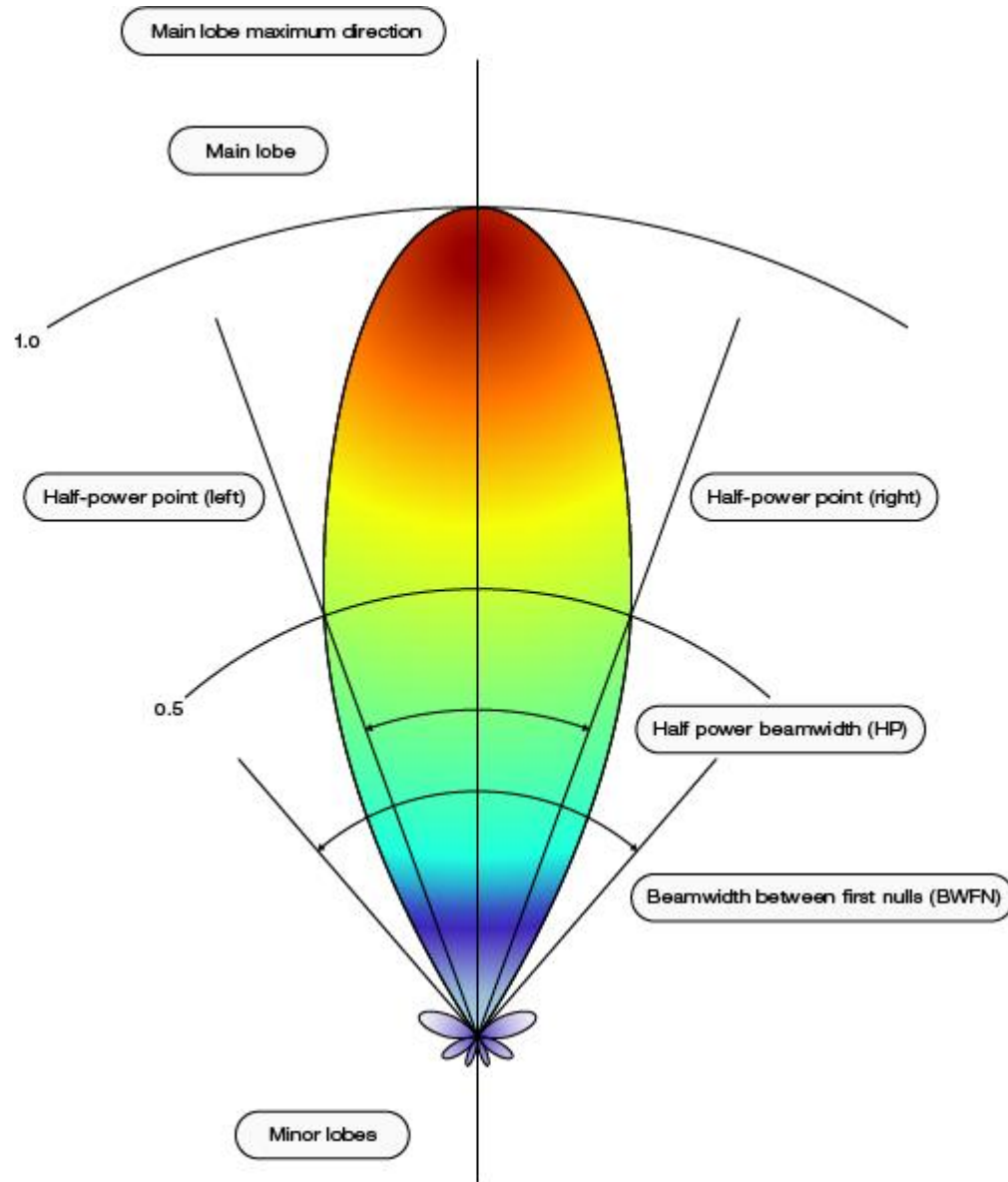
- Also called **half-power beam width**
- A measure of **directivity** of antenna.
- It is used to describe the resolution capabilities of the antenna to distinguish between two adjacent radiating sources or radar targets.



Radiation pattern in Cartesian coordinates



Cont...

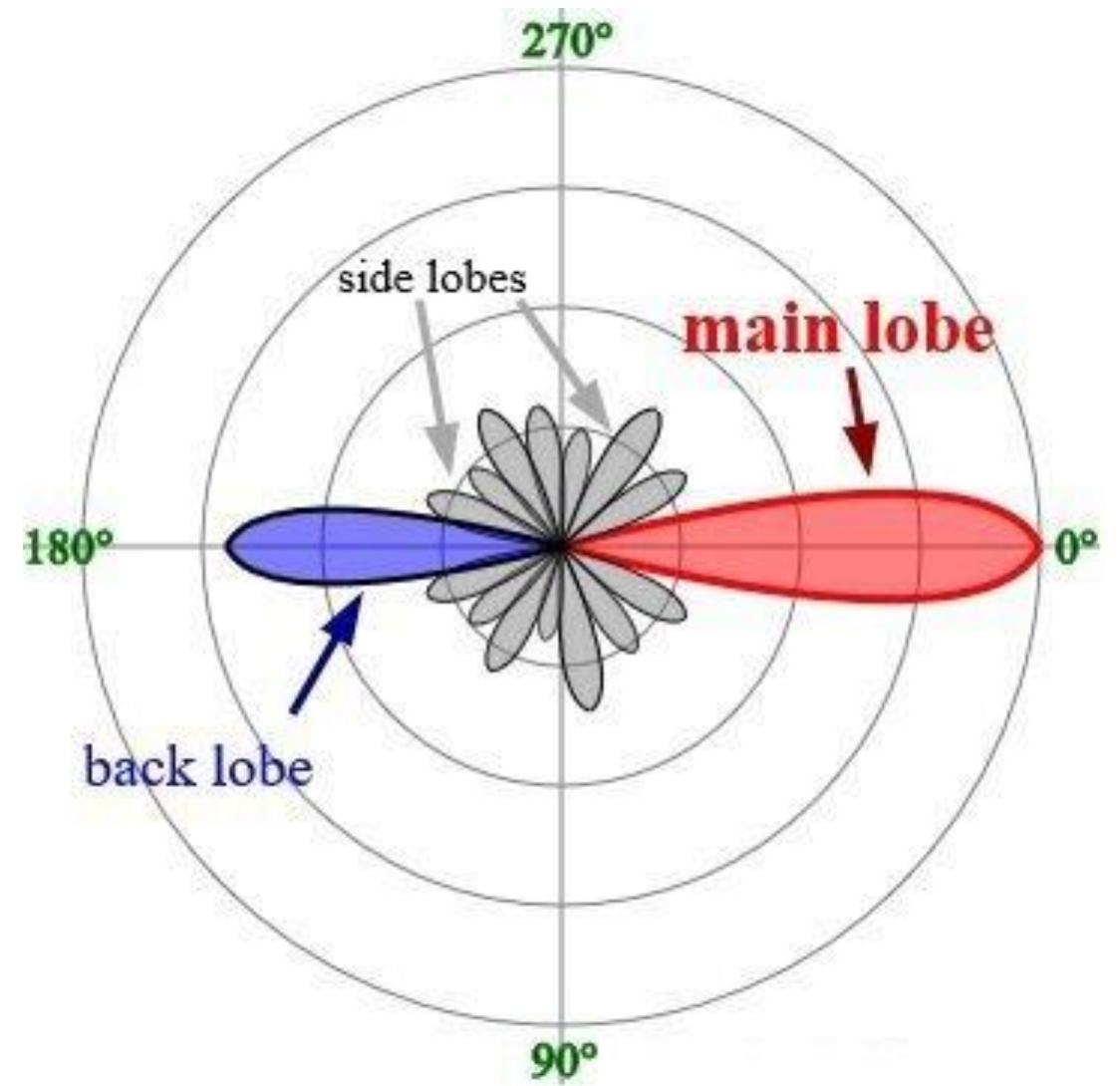


Main Lobe: Direction of maximum radiation.

Side Lobes: Undesirable, smaller peaks of radiation in other directions.

Nulls: Regions with almost no radiation.

Beamwidth: Commonly measured as the -3 dB (half-power) width to describe the main lobe's spread.



Radiation Pattern Parameters

- The **side lobe suppression** (or side lobe level) is a measure of the relation between the main lobe and the highest side lobe.

$$SLL = \frac{\text{Side Lobe Level}}{\text{Major Lobe Level}}$$

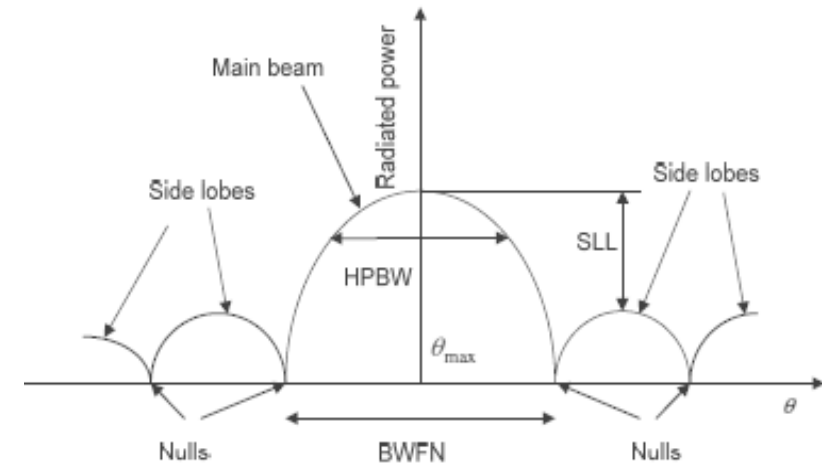
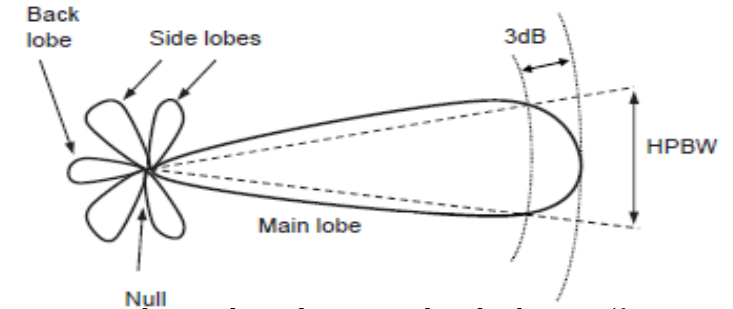
$SLL \cong -20 \text{ dB}$, not desirable

$SLL \cong -320 \text{ dB}$, desirable but difficult.

- The **half-power beamwidth (HPBW)** is the angle between the two points in the main lobe of an antenna pattern that are down from the maximum by 3 dB. It is usually defined for both principal plane patterns.

$$HP = |\theta_{HP\text{left}} - \theta_{HP\text{right}}|$$

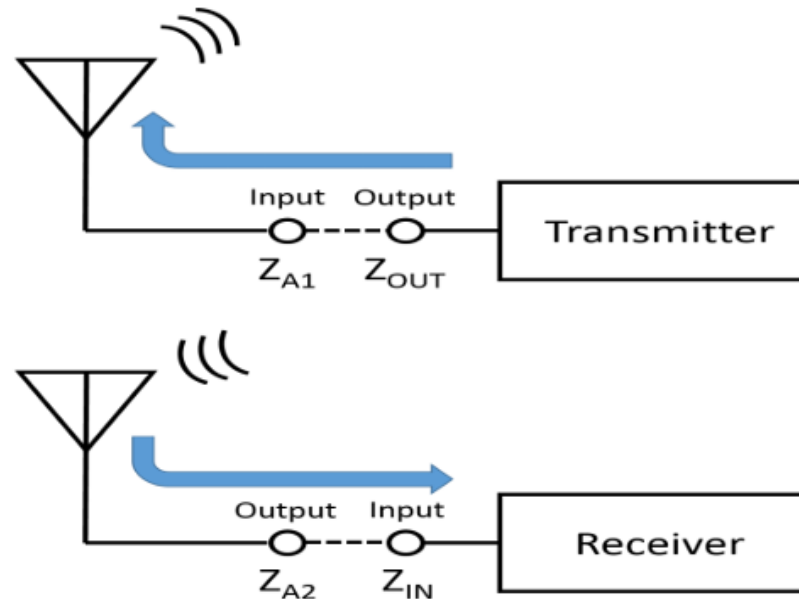
- The **front-to-back ratio** specifies the level of radiation from the back of a directional antenna. It is the ratio of the peak gain in forward direction to the gain in the reverse (180°) direction. It is usually expressed in dB



4. Antenna Impedance

- The antenna impedance pertains to the **transfer of power** from a generator to the antenna when the antenna is used as a transmitter and conversely, the transfer of power from the antenna to a load when the antenna is used as a receiver.
- For efficient transmission with no reflection, we should have: $Z_{A1} = Z_{OUT}$ and $Z_{A2} = Z_{IN}$

If the same antenna
then
 $Z_{A1} = Z_{A2}$

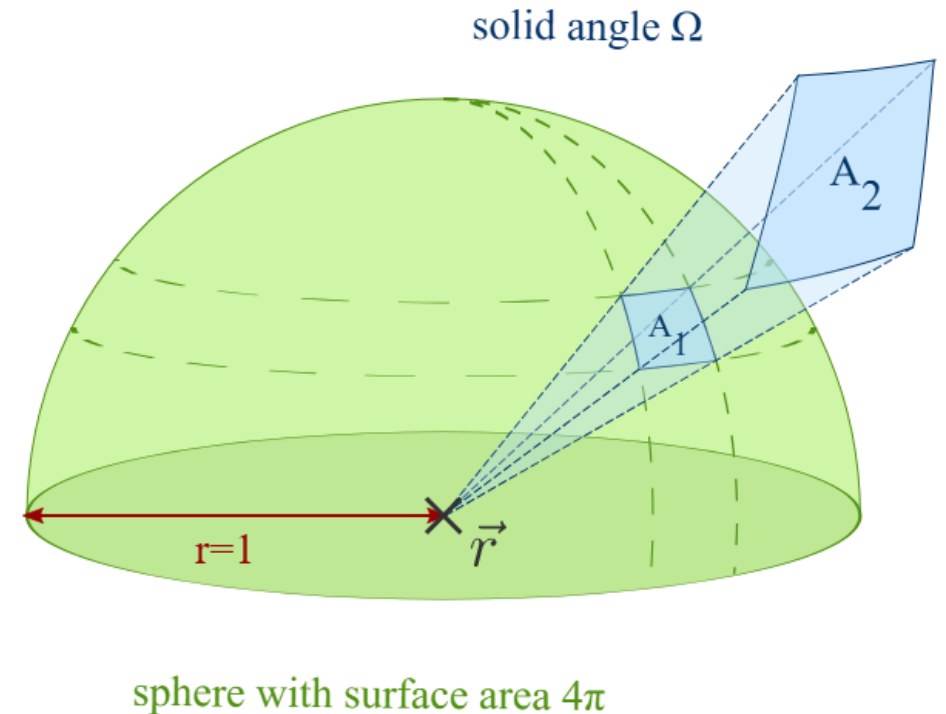


5. Radiation Intensity

- **Radiation intensity (U)** in a given direction is defined as the **power radiated** from an antenna **per unit solid angle**.
 - ✓ It is the **power radiated per unit solid angle**, measured in watts per steradian (**W/sr**).
 - ✓ It is basically the angular distribution of the radiated power around the antenna.
 - ✓ Crucial for calculating directivity

$$U = r^2 W_{rad}$$

- **One steradian** is defined as the **solid angle** with its vertex at the center of a sphere of radius r that is subtended by a spherical surface area equal to that of a square with each side of length r . Since the area of a sphere of radius r is $A = 4\pi r^2$, there are 4π sr in a closed sphere.



6. Gain

- The **gain** G is the **ratio** of the **radiation intensity** $U_{\max}$ obtained in the main direction of radiation to the **radiation intensity** U_{i0} that would be generated by a loss-free isotropic radiator with the same **input power** P_{t0} .
 - ✓ It is the ability of the antenna to radiate maximum power in a given direction (or) to absorb effectively
 - ✓ It describes how much power is transmitted in the direction of peak radiation to that of an isotropic source.

$$G = \frac{\text{Maximum radiation intensity from the subject antenna}}{\text{Maximum radiation intensity from the reference antenna}} = \frac{u_{\max}(\theta, \phi) \text{ of the subject antenna}}{u_{\max}(\theta, \phi) \text{ of the reference antenna}}$$

- In terms of the power gains

$$G = \frac{4\pi u_{\max}(\theta, \phi)}{W_t}$$

- where W_t is the total transmitted power.
- Not all transmitted power is radiated since there is a power loss W_L .
- $W_t = W_r + W_L$, where W_r is the radiated power.

$$G = \frac{4\pi u_{\max}(\theta, \phi)}{W_r + W_L}$$

7. Directivity (D)

- The **directivity factor D** is defined as the **ratio** of the **radiation intensity** $U_{\max}$ obtained in the main direction of radiation to the **radiation intensity** U_i that would be generated by a loss-free isotropic radiator with the same **radiated power** P_t .

✓ The extent to which an antenna concentrates its energy is given by the **directive gain**.

$$D = \frac{4\pi u_{\max}(\theta, \phi)}{W_r}$$

- The directivity D of an antenna is defined as the ratio of its maximum normalized radiation intensity, $F_{\max}$, to the average value of $F(\theta, \phi)$ over all directions (4π space):

$$D = \frac{F_{\max}}{F_{av}} = \frac{4\pi}{\Omega_p}$$

- The **radiation efficiency** is defined as:

$$\eta = \frac{P_{rad}}{P_t} \qquad G = \eta D$$

- For a lossless antenna, $\eta = 1$, and hence $G = D$.
- For an antenna with efficiency $\eta = 100\%$, this means that gain and directivity are equal.

8. Radiation Efficiency

- This is the **ratio of the power radiated** by the antenna to the **total input power** supplied to the antenna.

$$\text{Antenna efficiency } (\eta) = \frac{\text{Power radiated}}{\text{Total input power}} = \frac{P_{rad}}{P_{input}} = \frac{P_{rad}}{P_{rad} + P_L}$$

- P_{rad} =radiated power, P_L =power lost (ohmic losses)
- If the current flowing through the antenna is I , then:

$$P_{rad} = I^2 R_r$$

$$P_{loss} = I^2 R_L$$

- R_r =radiation resistance, R_L =ohmic (loss) resistance

$$\eta = \frac{I^2 R_r}{I^2 (R_r + R_L)} = \frac{R_r}{R_r + R_L}$$

- An ideal antenna is 100% efficient

9. Effective Area of a Receiving Antenna

- It is a **measure** for the **maximum received power** P_r that an antenna can pick up from a plane wave of the **power density** S :

- ✓ It is the area over which it extracts electromagnetic wave energy from the transmitted wave.
- ✓ The **effective area** A_w of an antenna is a parameter specially defined for receiving antennas.

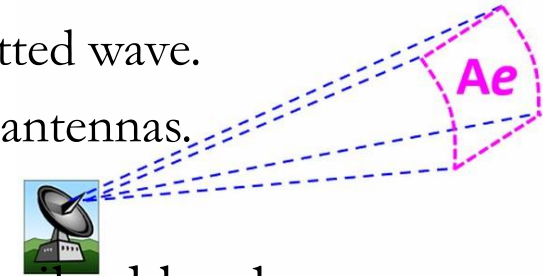
$$P_{r \max} = S \cdot A_w$$

- The relationship between the **effective** and the **geometrical areas** is described by the **aperture efficiency** q

$$q = \frac{A_w}{A_g}$$

- The **effective area** of an antenna can be converted to the **gain** and vice versa by means of the formula:

$$A_w = \frac{\lambda^2}{4\pi} G$$



10. Bandwidth of an Antenna

- The **bandwidth** of an antenna is defined as the **range of usable frequencies** - the range of frequencies over which the antenna can operate correctly.
 - ✓ The parameter most commonly taken into account here is the impedance match (i.e. VSWR < 2).
 - ✓ A ratio of 2:1 is called an **octave** - a ratio of 10:1 is a **decade**.
- It can also be described in terms of percentage of the center frequency of the band.

$$BW \text{ (in\%)} = \frac{(f_H - f_L)}{f_c} \cdot 100$$

where f_H is the highest usable frequency and f_L is the lowest usable frequency, and f_c is the center frequency.

Antenna Types

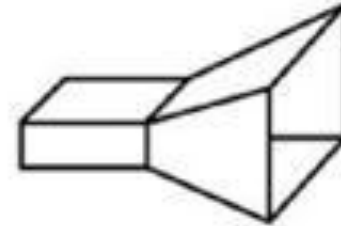
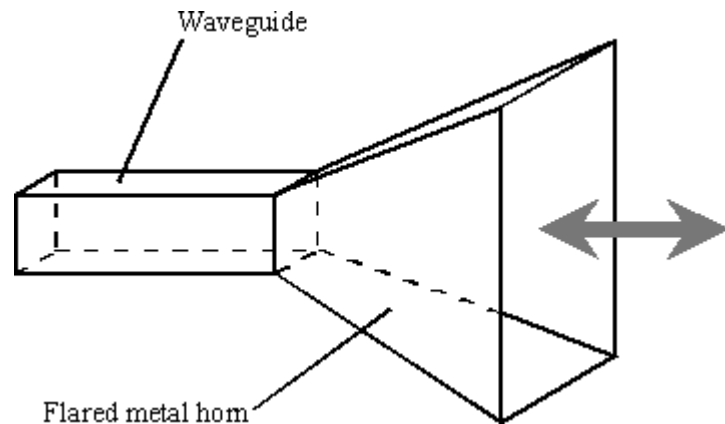
Wire Antennas

- ✓ These include **dipole**, monopole, helix, and loop antennas
- ✓ Usually used in personal applications, automobiles, buildings, ships, aircrafts and spacecrafts.

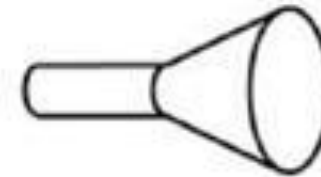


Aperture Antennas

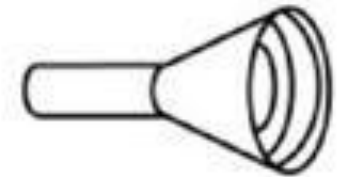
- Horn antennas, waveguide opening
- Usually used in aircrafts and space crafts, because these antennas can be flush-mounted



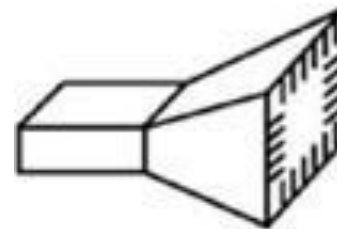
Simply flared rectangular horn



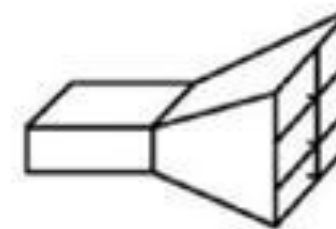
Simply flared conical horn



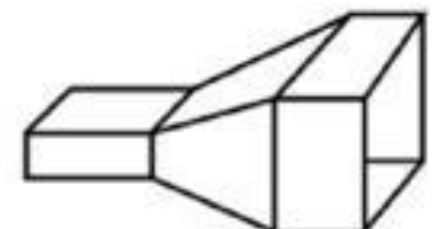
Corrugated conical horn



Finned horn



Segmented aperture horn



Compound flared multimode horn

Reflector Antennas

- **Parabolic** reflectors, corner reflectors
- These are high gain antennas usually used in radio astronomy, microwave communication and satellite tracking.
- The parabolic dish antenna consists of a small antenna at the focus of a large parabolic reflector, which focuses the signal in the same way as the reflector of a searchlight focuses a light beam.
- The beamwidth (at the 3 dB points) is approximately:

$$\theta = \frac{70 \lambda}{D}$$

- The gain is given by:

$$G = \eta \left(\frac{\pi D}{\lambda} \right)^2$$



Example

For a parabolic reflective antenna with a diameter of 2m, operating at 12GHz, with efficiency of 56%, what is the effective area and the antenna gain?

Solution

Wavelength $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{12 \times 10^9} = 0.025 \text{ m}$

Physical Area $A = \pi r^2 = \pi(1)^2 = \pi \text{ m}^2$

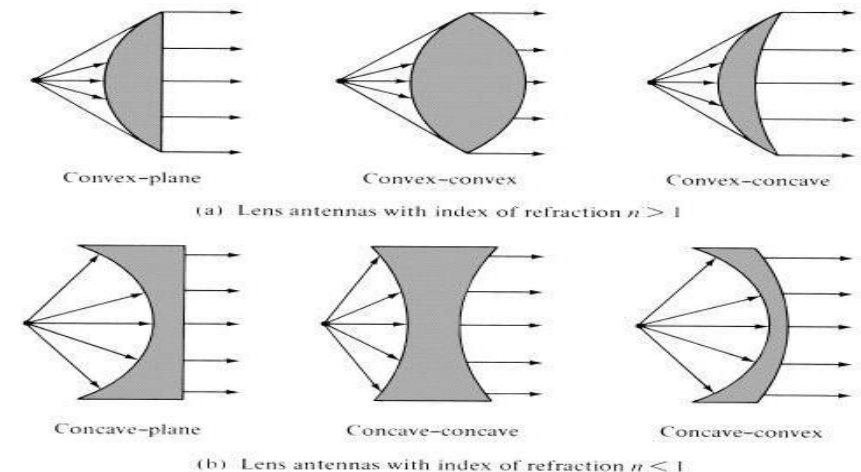
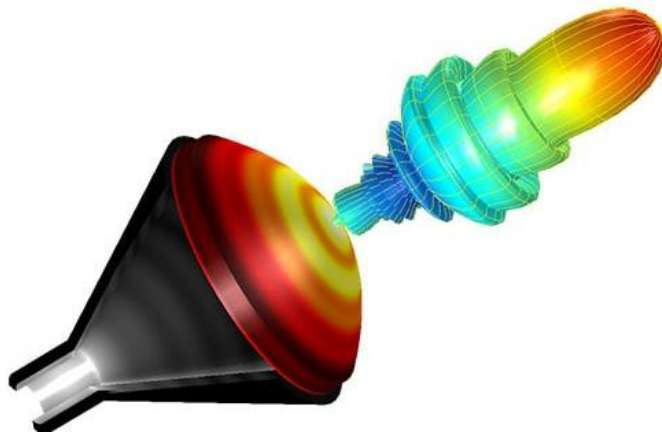
Effective Area $A_e = \eta A = 0.56\pi \approx 1.759 \text{ m}^2$

Antenna Gain $G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi \times 1.759}{(0.025)^2} = \frac{22.11}{0.000625} \approx 35376$

Gain in dB $G_{dB} = 10 \log_{10} 35376 \approx 45.5 \text{ dB}$

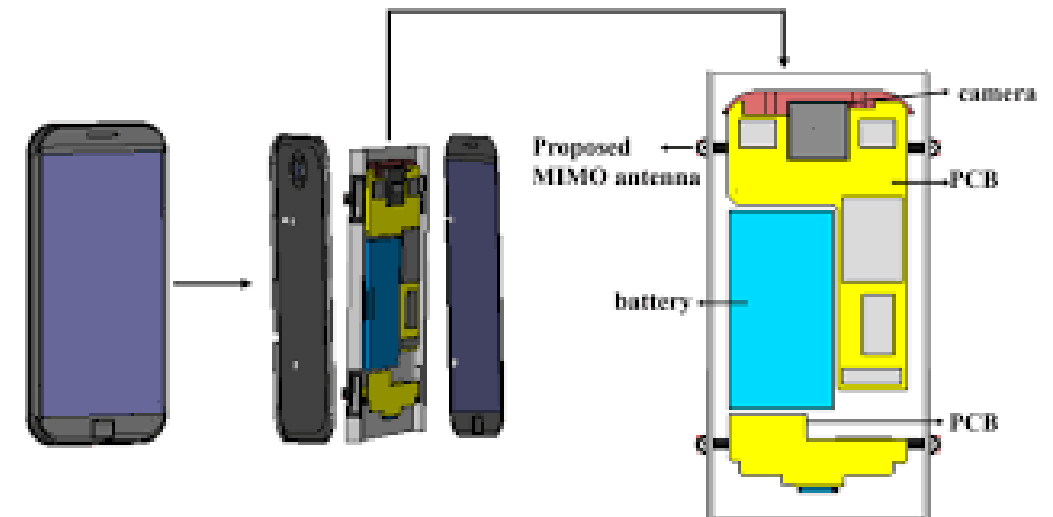
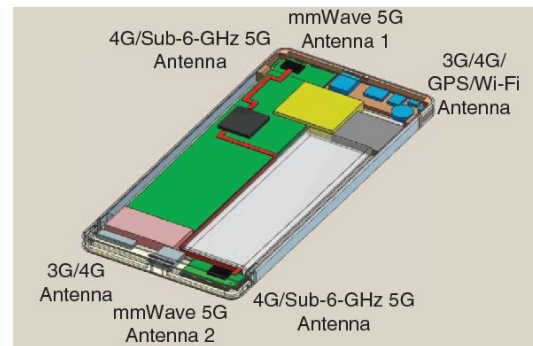
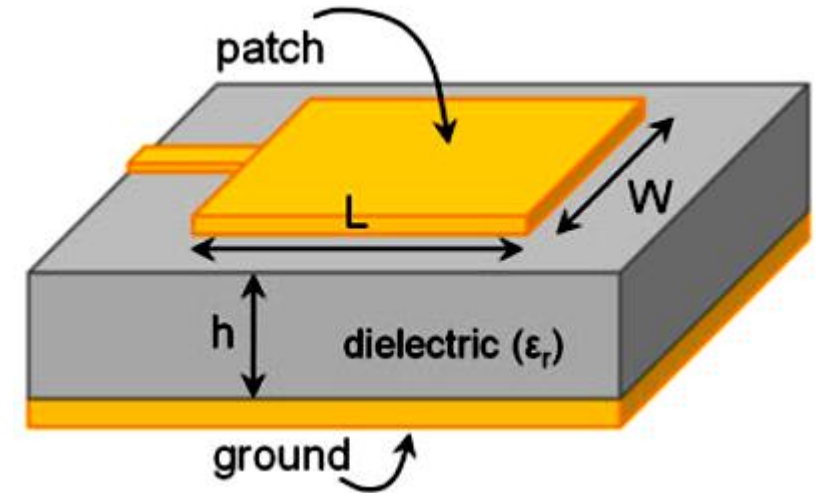
Lens Antennas

- Convex-plane, convex-convex, convex-concave and concave-plane lenses
- These antennas are usually used for very high frequency applications
- Used in radar systems, wireless communication systems, and other applications where precise control of the electromagnetic radiation is required.



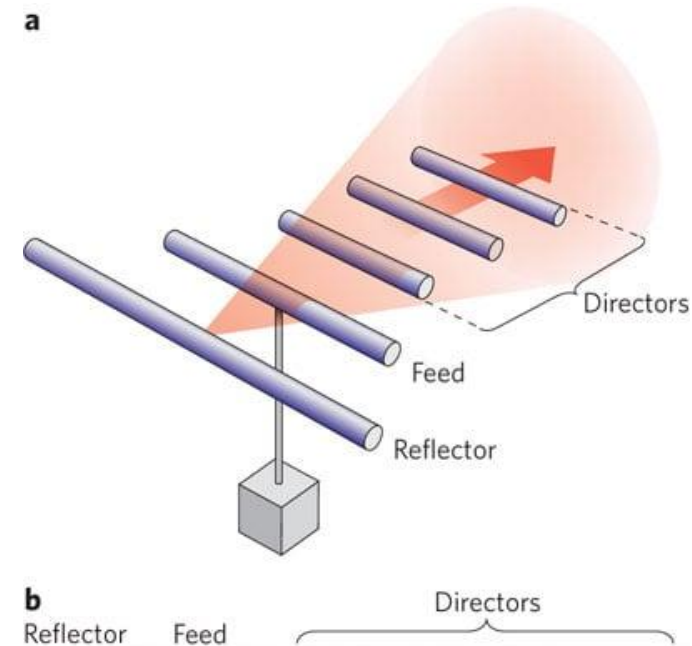
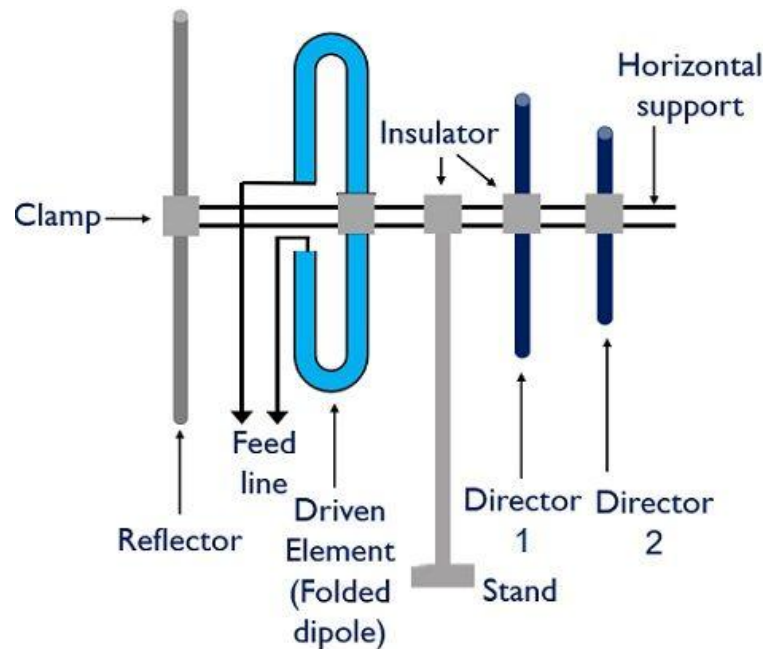
Microstrip Antennas

- Low-profile antenna consisting of a conducting patch printed on one side of a dielectric substrate, with a ground plane on the other side. Rectangular, circular etc. shaped metallic patch above a ground plane
- Widely used in modern wireless systems because of their compact size and ease of fabrication.
 - ✓ Used in aircraft, spacecraft, satellites, missiles, cars, mobile phones etc.



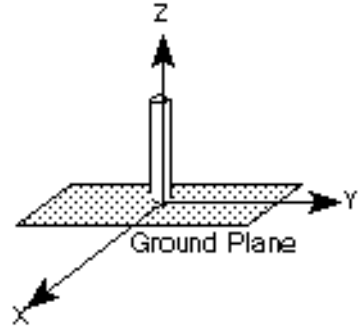
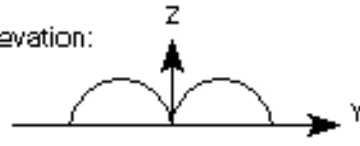
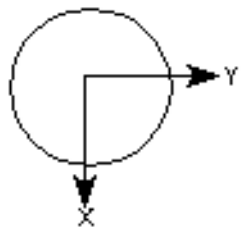
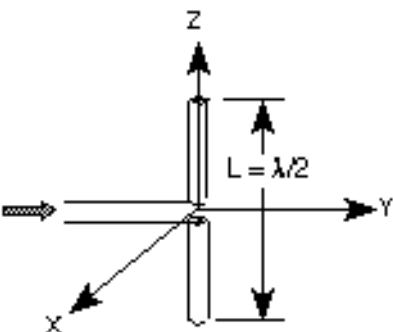
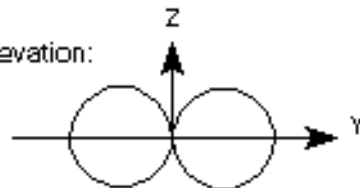
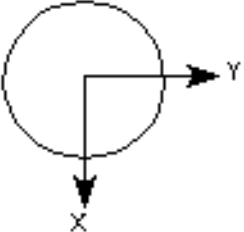
Array Antennas

- **Yagi-Uda** antenna, microstrip patch array, aperture array, slotted waveguide array.
- Used for very high gain applications with added advantage, such as, controllable radiation pattern.



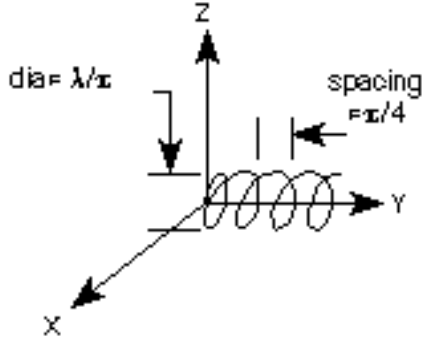
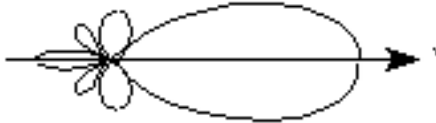
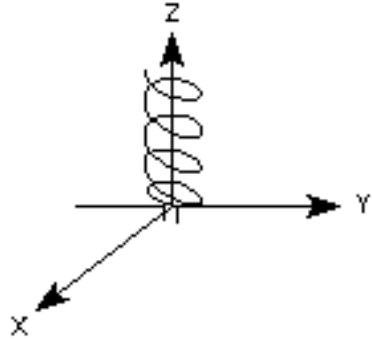
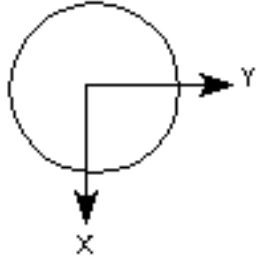
Antenna Specifications

Pole Antennas

Antenna Type	Radiation Pattern	Characteristics
MONOPOLE 	Elevation:  Azimuth: 	Polarization: Linear Vertical as shown Typical Half-Power Beamwidth 45 deg x 360 deg Typical Gain: 2-6 dB at best Bandwidth: 10% or 1.1:1 Frequency Limit Lower: None Upper: None Remarks: Polarization changes to horizontal if rotated to horizontal
$\lambda/2$ DIPOLE 	Elevation:  Azimuth: 	Polarization: Linear Vertical as shown Typical Half-Power Beamwidth 80 deg x 360 deg Typical Gain: 2 dB Bandwidth: 10% or 1.1:1 Frequency Limit Lower: None Upper: 8 GHz (practical limit) Remarks: Pattern and lobing changes significantly with L/f. Used as a gain reference < 2 GHz.

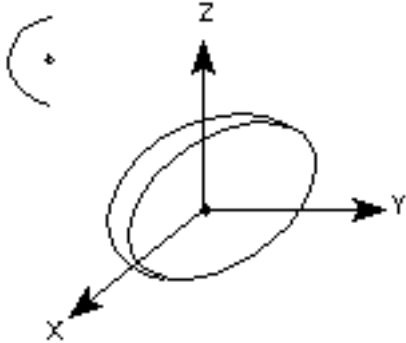
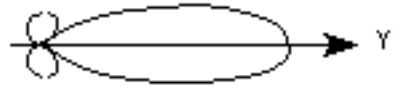
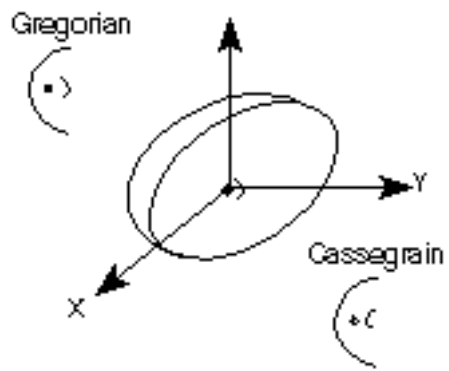
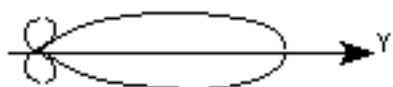
Antenna Specifications

Helix Antennas

Antenna Type	Radiation Pattern	Characteristics
AXIAL MODE HELIX 	Elevation & Azimuth 	Polarization: Circular Left hand as shown Typical Half-Power Beamwidth: 50 deg x 50 deg Typical Gain: 10 dB Bandwidth: 52% or 1.7:1 Frequency Limit Lower: 100 MHz Upper: 3 GHz Remarks: Number of loops >3
NORMAL MODE HELIX 	Elevation:  Azimuth: 	Polarization: Circular - with an ideal pitch to diameter ratio. Typical Half-Power Beamwidth: 60 deg x 360 deg Typical Gain: 0 dB Bandwidth: 5% or 1.05:1 Frequency Limit Lower: 100 MHz Upper: 3 GHz

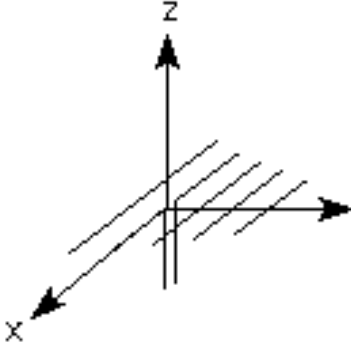
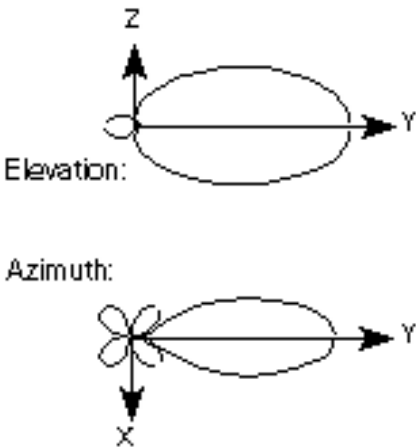
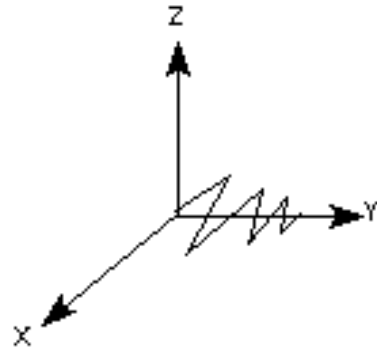
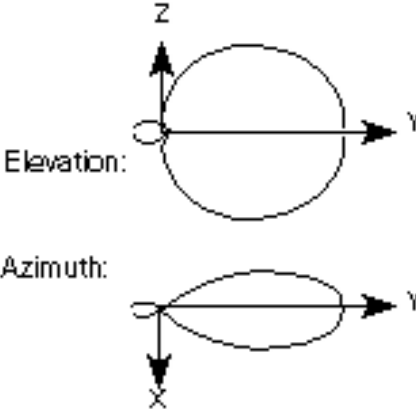
Antenna Specifications

Parabolic Antennas

Antenna Type	Radiation Pattern	Characteristics
PARABOLIC (Prime) 		Polarization: Takes polarization of feed Typical Half-Power Beamwidth: 1 to 10 deg Typical Gain: 20 to 30 dB Bandwidth: 33% or 1.4:1 limited mostly by feed Frequency Limit: Lower: 400 MHz Upper: 13+ GHz
PARABOLIC Gregorian  Cassegrain 		Polarization: Takes polarization of feed Typical Half-Power Beamwidth: 1 to 10 deg Typical Gain: 20 to 30 dB Bandwidth: 33% or 1.4:1 Frequency Limit: Lower: 400 MHz Upper: 13+ GHz

Antenna Specifications

Yagi Uda Antennas

Antenna Type	Radiation Pattern	Characteristics
YAGI 	 Elevation: Azimuth:	Polarization: Linear Horizontal as shown Typical Half-Power Beamwidth 50 deg X 50 deg Typical Gain: 5 to 15 dB Bandwidth: 5% or 1.05:1 Frequency Limit Lower: 50 MHz Upper: 2 GHz
LOG PERIODIC 	 Elevation: Azimuth:	Polarization: Linear Typical Half-Power Beamwidth: 60 deg x 80 deg Typical Gain: 6 to 8 dB Bandwidth: 163% or 10:1 Frequency Limit Lower: 3 MHz Upper: 18 GHz Remarks: This array may be formed with many shapes including dipoles or toothed arrays.